

Data Base Management System

1. Self-describing nature of DB.
Catalog $\rightarrow$ Structure of the files, database.

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$\rightarrow$ metadata (data about the database)

2. Isolation between Programs, Data and Data abstraction.

$\rightarrow$ Data abstraction does not give information about the structure of data, way to access the data.

$\rightarrow$ Programs operate on data by certain operations through name, arguments, etc. without knowing info. about the implementation of operation.

$\rightarrow$ Program-data independence + Program-operation independence

Data Abstraction
Data Model

Type of data abstraction used for conceptual representation of the data

3. Support of Multiple views of the data:-
 $\rightarrow$ Different types of users, perspective or view of database
 $\rightarrow$ View is a subset of database
 $\rightarrow$ virtual data that is desired but not stored.

4. Sharing of Data and Multiuser Transaction Processing:-
 $\rightarrow$ Allows multiple users to access the database at same time
 $\rightarrow$ DBMS must have concurrency control trying to update the same data in controlled manner such that result of update is correct. $\rightarrow$ Online Transaction Processing (OLTP).

Data warehouse:- Storing a huge voluminous data

Extract Transform Load (ETL) to extract certain data from the data warehouse and store it in data marts.

Data Warehouse $\xrightarrow{\text{ETL}}$ Data Marts $\rightarrow$ Database

Transaction is a process that has 1 or more databases accesses like reading / writing / etc. updating.

$\rightarrow$ Atomic

source

$\rightarrow$ Isolation

other trans

$\rightarrow$ Consistency

$\rightarrow$

$\rightarrow$ Data and its

understand

$\rightarrow$ Data

describe

$\rightarrow$ Structure

constraints

(i). High

the way

(ii). Low

is its

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can

info.

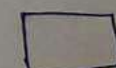
$\rightarrow$ Entity

$\rightarrow$ Attribute

$\rightarrow$ Relationship

among

$\rightarrow$ Data



FOOD COL

→ Atomicity :- Either transaction is executed or not at all
[ACID]

→ Isolation :- Each transaction executes in isolation from other transactions, even though they are concurrently executed.

→ Consistency :- ✓

→ :-

Data Models, Schemas, Instance

→ Data abstraction :- Suppression of details of data organization and storage, highlighting the essential features for improved understanding of data.

→ Data Model :- Collection of concepts that is used to describe the structure of the database.

→ Structure of database :- Has data types, relationships and constraints

Category of Data Model :-

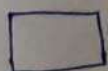
- (i). High level data model → Provides concepts that are close to the way many users perceive the data.
- (ii). Low level / Physical level → Provides concepts on how data is stored on computer storage media.
- (iii). Representational / Implementation level → Provides concepts that can be understood by end-user and also has certain info. about data storage.

Entity

- Entity is a real world object that describes it.
- Attribute is a property of an entity represented on association.
- Relationship among 2 or more entities.

→ Database schema :- Description of the database / structure of DB.

ER diagram



Entity

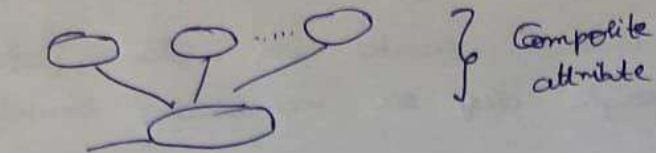


Weak Entity

→ dependent on the strong entity

[Don't have existence on either own]

Attribute [of an entity]



[Eg: First

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Address attribute



Derived attribute [Eg: Age obtained from DOB]



Multivalued attribute

[Eg: degree | education, vehicle type]



key attribute

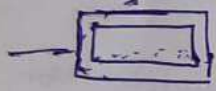
[Eg: USN, unique value associated]



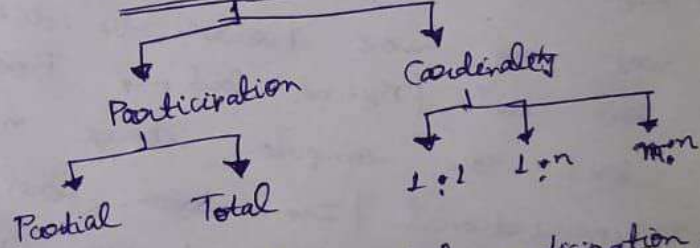
relationship

[Connection between 2 or more attribute entities]

- Strong entities can be identified with key attribute
- Weak entities can be identified with partial key

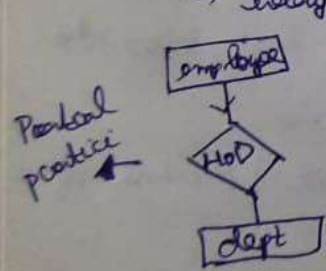


Structural constraints.

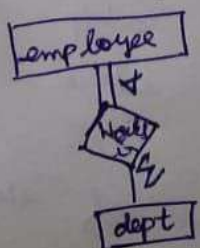


→ In ER diagram — (single line) partial participation.

Eg: In RVC, in employee table, department table, ⇒ every employee belongs to any of the dept.



Partial participation

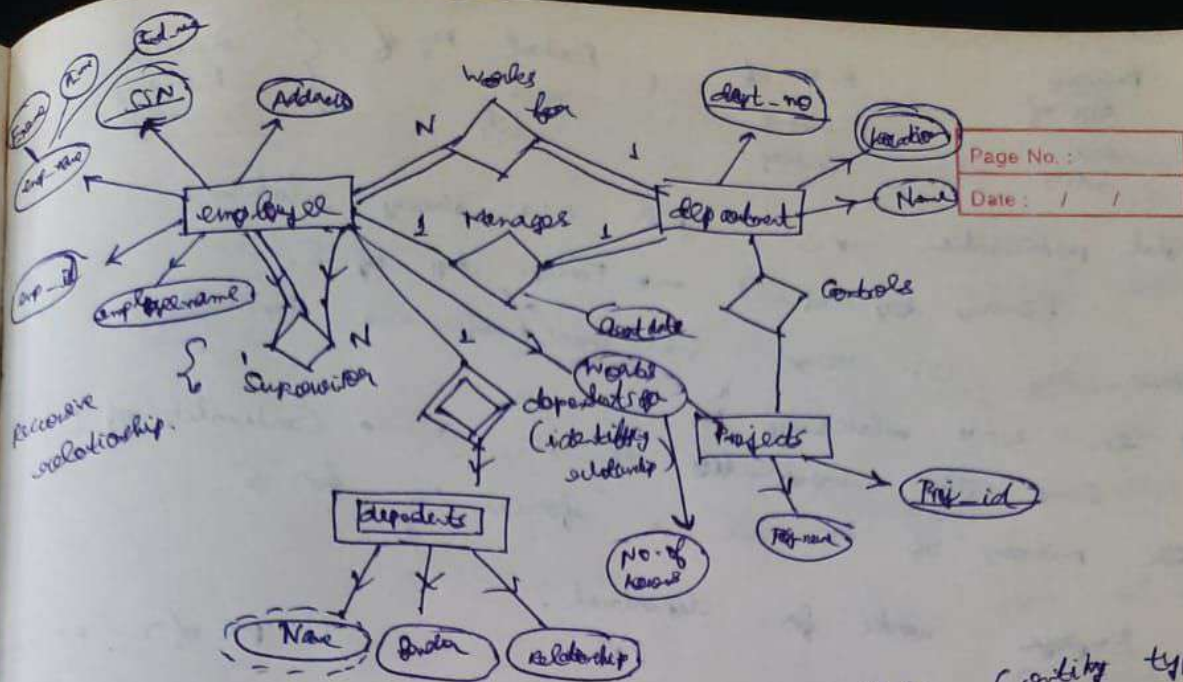


→ Full participation.

|| → For all every.

| → atleast one

all nouns → entities verbs → relationship.



- Set of entities which has same attributes (entity type).
- Binary Relationship: Relation between 2 entities
- That entity always has total participation constraint.

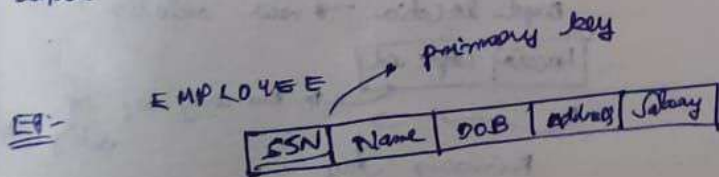
Relational model

→ In E-R diagram, it is called entity, attribute, in relational model, it is called as relational schema.

NULL → value unknown, Not available, Attribute does not apply.

ER to Relational Mapping

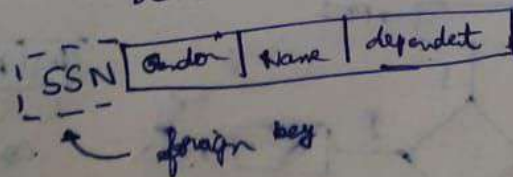
For Entity E in ER schema, create a relation which has same attributes



DEPARTMENT

<u>Dept-id</u>	<u>Dept-Name</u>	<u>Location</u>
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→ For weak entities :-
DEPENDENT



Owner entity (Employee's)
primary key is connected
to dependent (weak entity)
as foreign key.

→ For weak entity, the primary key is combination of
primary key of strong entity (Employee) and partial key
of weak entity.

Primary key of weak entity

P.K. of strong entity

Partial key of weak entity

Composite Primary Key

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Total participation $\rightarrow S$

For 1:1 binary relationship.

Primary key of T $\rightarrow$ Foreign key of S.

(i) Foreign key (ii) Merge (iii) Cross referencing approach.

$\rightarrow$ In 1-N relationship R,

S $\rightarrow$ the cardinality of N. T $\rightarrow$ Cardinality of 1.

The primary of T will be foreign key for S.

Employee works for department.

(1)

PK of T = FK of S

(N)

$\rightarrow S$

SSN | Name | DOB | ... | EMPLOYEE

Dept_id

Dept_id | Nonlocal SSN

$\rightarrow$ PK of department

Cross reference $\rightarrow$ Make separate relation using PK of S and T.

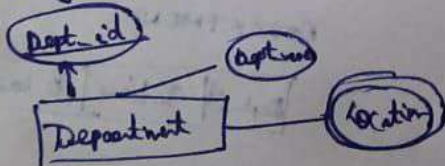
Dept_id | SSN } New Relation

Multivalued attribute mapping:-

For each multivalued attribute 'A' create new relation R.

This new 'R' will have the attribute 'A' + primary key of the entity which had this attribute 'A' as foreign key.

Eg:-



dept. location $\rightarrow$ new relation R.

Location | Dept_id

Binary key of entity (dept)

Primary key (Composite)

N-ary Relationship:-

Relation between

≥ 2 entities.

PK1



PK2

PK3

PK1 | PK2 | PK3 | ...

Composite primary key

Simple attributes.

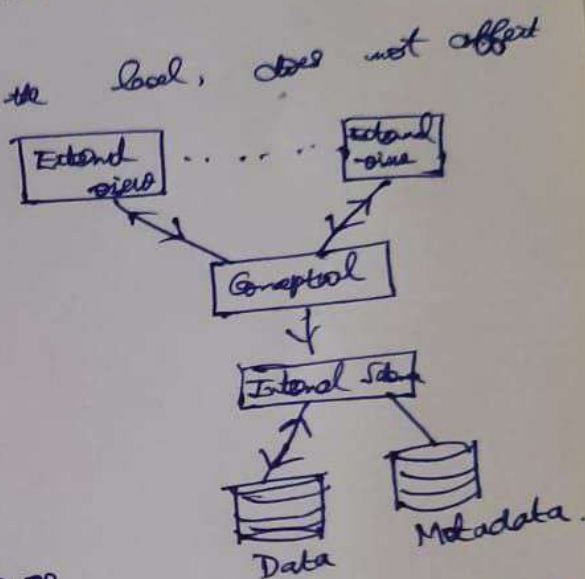
States in DB:-

↳ Empty state (No entries) → Initial state (Initial set of entries) → Current state (Current state of DB) → Valid state (Satisfies all constraints).
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3 schema architecture:-

- (i) Internal schema → Info. about physical storage structure of the DB. [Physical data model]
- (ii) Conceptual schema → Structure of whole DB (entities, attributes, relations, constraints, data types) High-level data model.
- (iii) External schema → Focus on user perspective depending on user perspective (view level).

Mapping: changes made in any of the lower layers.



Data independence:-

↳ Change in one schema level should not affect higher levels.

① Logical data independence:- Capacity to change conceptual schema without

changing external schema.

(Adding rows / entries does not change the view in user's ad).

(ii) Physical data independence:- Capacity to change internal schema without changing conceptual schema.